

Cloud service provider	Options and capabilities for training	Options and capabilities for accuracy and precision	Options and capabilities for NLP
Microsoft	Advanced and flexible options and capabilities for training, using various methods and modes, such as supervised, unsupervised, semi-supervised, and reinforcement learning, as well as various degrees of automation, such as manual, automated, and hybrid.	A limited and nascent portfolio of options and capabilities for providing and ensuring high levels of accuracy and precision, mainly focused on data and model management, using various types of data and information, such as text, speech, image, video, and audio.	A unified and simplified way of building, deploying, and managing large language models and natural language processing technologies and services, such as AutoML, AI Platform Text, AI Platform Speech, and Cloud Translation, which can generate and process various types of natural language, such as text and speech.
Amazon	A moderate and diverse portfolio of options and capabilities for training, using mainly supervised and unsupervised learning, as well as some degree of automation, such as semi-automated and fully automated.	A comprehensive and mature portfolio of options and capabilities for providing and ensuring high levels of accuracy and precision, using various types of data and information, such as text, speech, image, video, and audio, as well as various methods and techniques, such as data labelling, data quality, data augmentation, and model testing.	A fragmented and complex way of building, deploying, and managing large language models and natural language processing technologies and services, such as Amazon Comprehend, Amazon Polly, Amazon Transcribe, and Amazon Translate, which can generate and process various types of natural language, such as text and speech, but require more integration and configuration efforts.
Google	A limited and narrow portfolio of options and capabilities for training, using mainly supervised learning, but with a high degree of automation.	A moderate and emerging portfolio of options and capabilities for providing and ensuring high levels of accuracy and precision, using mainly text and speech data and information, as well as some methods and techniques, such as data labelling, data quality, and model testing.	A comprehensive and advanced way of building, deploying, and managing large language models and natural language processing technologies and services, such as Google Cloud AutoML Natural Language, Google Cloud Natural Language API, Google Cloud Speech-to-Text, and Google Cloud Translation API, which are capable of generating and processing various types of natural language, such as text and speech, but require more technical and domain expertise.

Table 1: Industrial cloud solutions offered by the three major cloud service providers

operational efficiency of industrial companies.

Reduced infrastructure costs

- **Hardware savings:** By moving to the cloud, companies can avoid the substantial upfront costs associated with purchasing and maintaining hardware. Cloud providers manage the infrastructure, reducing the need for on-premises servers and data centres, which cuts down on power costs and the requirement for physical space.

- **Lower maintenance costs:** Cloud providers handle maintenance, repairs, and upgrades, freeing up in-house IT staff to focus on more strategic initiatives rather than routine maintenance tasks.

Enhanced scalability at lower costs

- **Pay-as-you-go model:** Cloud computing offers a pay-as-you-go pricing structure, allowing companies to pay for services only when they are used. This model eliminates the need for

overprovisioning and reduces costs associated with unused capacity.

- **Scalability:** Companies can quickly scale up or down to meet changing business demands without the need for significant capital investments in hardware or software.

Labour efficiency

- **Reduced IT staff:** With cloud providers managing the infrastructure, companies can reduce their IT staff size or reallocate personnel to more valuable tasks,